

Mozammal Hossain

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WORK EXPERIENCE

IQVIA

A Fortune 500 global leader in healthcare analytics and technology

Software Development Engineer II

Mar 2026 - Present

Received IQVIA Impact Award

- Contributed to the broad industry adoption of IQVIA's agent solutions, now embedded in the workflows of 19 of the top 20 global pharmaceutical companies, underscoring strong market trust in IQVIA's AI capabilities.
- Architected an enterprise-grade agent memory system (STM LTM) for the core agent platform, significantly improving context retention, personalization, and scalability.
- Integrated the A2A protocol into the existing LangGraph ecosystem, enabling flexible, standards-based orchestration of sub-agents that can be reused across diverse platforms.
- Introduced and productionized deep agents within the current infrastructure to support long-running, in-depth research workflows, expanding the platform's ability to handle multi-stage analytical tasks.
- Demonstrated strong ownership and accountability for key independent agent platforms, including GMI Agent and Field Force Agent, ensuring reliability, continuous improvement, and alignment with business needs.

Software Development Engineer I

Mar 2024 - Mar 2026

- Build AI-driven solutions as part of the AI Assistant team within the Orchestrated Analytics Department, enabling real-time, data-driven decision-making for pharmaceutical commercialization and life sciences workflows.
- Drive end-to-end AI workflows for IQVIA's next-generation data exploration and orchestration agent platform, powering scalable analytics and insights across commercial pharma use cases.
- Design and develop advanced multi-agent backend systems with modular, scalable architectures to support enterprise-grade AI Assistant capabilities.
- Implement and optimize advanced reasoning techniques, engineering robust tool-calling frameworks—including MCP, context engineering, skill orchestration, and tool interception—to enhance agent intelligence and performance.
- Architect and optimize AI agents operating on petabyte-scale pharmaceutical datasets in Snowflake, enabling efficient querying, deep reasoning, and real-time insight generation across complex healthcare and life sciences data ecosystems.

DVIZ TECHNOLOGIES, DELAWARE, USA

Software Engineer (Remote)

July 2023 - Feb 2024

- Led the development of backend solutions for agentic Retrieval-Augmented Generation (RAG) applications with fine-tuned LLMs.
- Built and optimised two AI-powered MVPs for data-driven decision-making platforms.
- Designed a statistical model and performed time-series data analysis for a fintech solution, enabling predictive analytics and actionable insights.

RESEARCHES

DEEP LEARNING IN PROSTATE CANCER DIAGNOSIS AND GLEASON GRADING IN HISTOPATHOLOGY IMAGES(PUBLISHED).

- Explored Deep Learning for prostate cancer diagnosis in histopathology.
- Analyzed image pre/post-processing, evaluation techniques, and existing deep learning approaches for differentiating Gleason grades.
- Identified promising solutions and discussed limitations and future scope.

PROJECTS

A full-stack community-based networking web app to facilitate connections between current undergraduate students and alumni.

- Built real-time messaging and notification system using **Socket.io**.
- Integrated **Cloudinary** as a **CDN** to serve / store media content fast and reduce loading time also used **presigned** URLs to control access.

Tools/Tech: Next.js, Express.js, Socket.io, MongoDB, Sendgrid

EDUCATION

SHAHJALAL UNIVERSITY OF

SCIENCE AND TECHNOLOGY

BSc in Computer Science and

Engineering

CGPA - 3.60

SKILLS

PROGRAMMING (Solved 600+)

[LeetCode](#) | [HackerRank](#) | [Codeforces](#) | [VJudge](#) | [LightOJ](#)

LANGUAGE

Python | JavaScript | TypeScript |

Java | C | C++ | Bash | SQL

TOOLS/TECH

FastAPI | Django | DRF | Celery | REST | NextJS | NodeJS | Express | Git | GraphQL | Socket.io | Docker | K8S | Redis | RabbitMQ

DATABASE

MySQL | PostgreSQL | MongoDB | VectorDBs | Neo4j

CLOUD (AWS)

AWS Lambda | S3 | EC2 | Glue | Amazon RDS | SageMaker | Bedrock EKS | CI/CD

AI TOOLS/TECH

LangGraph | LangChain | LangSmith | LlamaIndex | MCP | A2A | vLLM | n8n | ADK | Vertex AI Agent Engine | PyTorch | Keras | Scikit-Learn | Matplotlib | NumPy | Pandas

DATA ENGINEERING

TOOLS/TECH

ETL | ELT | Scrapy | PySpark | Airflow Kafka

ACTIVITIES

- Conducted **workshops** on **Machine Learning and Deep Learning**.
- Participated in the Bangladesh Physics Olympiad 2017 at the national level.